

Product Requirements

Accessible Interview System Project

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Brief problem statement

Job seekers, especially those who experience social anxiety or are on the autism spectrum, may find job interviews stressful and may have limited opportunities to practice in a low-pressure environment. Traditional interview practices may not provide enough flexibility or accessibility for users who need to practice at their own pace or receive clear feedback about their answers.

The Accessible Interview System provides a low-pressure environment where job seekers can practice technical interviews using an AI-powered interviewer. The system can generate technical interview questions based on a user's job description or requirements and allow the user to practice answering them. After an interview, the system provides feedback and performance information to help users identify areas for improvement and track their interview practice.

System requirements

The deployed system shall require the following:

- **SR1:** The user shall have access to a modern web browser, such as a current version of Google Chrome, Microsoft Edge, Firefox, or Safari.
- **SR2:** The user shall have a reliable internet connection to access the web application and its online services.
- **SR3:** The system shall be hosted on a server capable of running the application's frontend and backend services.
- **SR4:** The deployed system shall have access to the application's MySQL database for storing user accounts, interview responses, feedback, and progress data.
- **SR5:** The deployed system shall have access to the OpenAI API for generating and conducting AI-based technical interview questions and responses.
- **SR6:** The system shall support access from computers, tablets, and mobile devices through a web browser.
- **SR7:** The system shall support text-to-speech functionality for users who choose to display text read aloud.

Users profile

Primary User Profile: Job Seekers Preparing for Technical Interviews

Who are they:

Job seekers, including students and recent graduates, who are preparing for technical interviews in software engineering, computer science, information technology, or related fields.

Reading Level:

Users should be able to read and understand general written instructions and technical interview questions.

Relevant Traits:

Users may experience social anxiety or may be on the autism spectrum. The system is designed to provide a low-pressure, repeatable, and self-paced environment for interview practice.

Accessibility Needs:

Users may benefit from keyboard navigation, screen-reader support, adjustable text display, and text-to-speech functionality. The interface should avoid unnecessary flashing or distracting visual elements.

Technical Proficiency:

Users should be able to navigate a modern web browser and interact with basic web applications.

List of Features

1. F1 – Interview Question Library:

The system shall provide users with a collection of technical interview questions

relevant to software engineering, computer science, information technology, and related positions.

2. **F2 – Job Description-Based Questions:**
The system shall generate technical interview questions based on the job description or requirements provided by the user.
3. **F3 – Text-Based Interview:**
The system shall allow users to participate in technical mock interviews using text-based communication.
4. **F4 – AI Simulated Interview:**
The system shall allow users to practice technical interviews with an AI-powered interviewer that presents questions and responds to the user's answers.
5. **F5 – Interview Feedback:**
The system shall provide feedback after an interview that identifies strengths and areas for improvement in the user's responses.
6. **F6 – User Answer Analysis:**
The system shall analyze the user's interview answers and provide information about the content and quality of the responses.
7. **F7 – Interview Report:**
The system shall generate a report summarizing the user's interview performance and areas that may require additional practice.
8. **F8 – One-Off Practice Questions:**
The system shall allow users to practice individual technical interview questions without completing a full mock interview.
9. **F9 – Pause and Resume:**
The system shall allow users to pause a simulated interview and resume it later during the same interview session.
10. **F10 – Accessibility and Visual Adjustments:**
The system shall provide accessibility options that allow users to adjust the presentation of text and other interface elements to better fit their needs.
11. **F11 – Screen Reader Support:**
The system shall support screen-reader access for navigation and interview content.
12. **F12 – Text-to-Speech:**
The system shall allow users to have supported displayed text converted into spoken audio.

13. F13 – User Statistics:

The system shall provide performance statistics that allow users to identify questions or areas in which they need additional practice.

14. F14 – Interview History and Progress:

The system shall allow users to review previous interview sessions and track their practice progress over time.

15. F15 – User Account Management:

The system shall allow users to create an account, log in, and **log out**.

Functional requirements (user stories)

List the Priority as 1 (High Priority - Critical) to 3 (Low Priority – Would be nice if we have time)

No.	User Story Name	Description	Priority
R1	Create Account	As a user, I want to create an account so that I can save my interview sessions and track my progress.	1
R2	Log In	As a user, I want to log in to my account so that I can access my saved interview data and progress.	1
R3	Select Technical Interview	As a user, I want to select a technical interview so that I can practice questions relevant to technical jobs.	1
R4	Start Mock Interview	As a user, I want to start a technical mock interview so that I can practice answering interview questions.	1
R5	Generate Interview Questions	As a user, I want the system to generate technical interview questions based on my job description so that I can practice questions relevant to the position.	1

R6	Answer Interview Questions	As a user, I want to submit answers to interview questions so that I can practice responding to technical questions.	1
R7	Analyze Answers	As a user, I want the system to analyze my interview answers so that I can receive feedback about the content and quality of my responses.	1
R8	Receive Interview Feedback	As a user, I want to receive feedback after an interview so that I can identify my strengths and areas for improvement.	1
R9	Review Interview Report	As a user, I want to view an interview report so that I can review my overall performance and areas that need improvement.	1
R10	Practice Individual Questions	As a user, I want to practice individual technical interview questions so that I can work on specific areas without completing a full interview.	2
R11	Pause Interview	As a user, I want to pause and resume a mock interview so that I can take a break when needed.	2
R12	Adjust Visual Settings	As a user, I want to adjust how text is displayed so that I can read the interface in a way that better fits my needs.	2
R13	Use Text-to-Speech	As a user, I want to use text-to-speech so that I can hear supported text instead of reading it.	2

R14	View Statistics	As a user, I want to view my interview performance statistics so that I can identify questions or areas where I need more practice.	2
R15	Track Progress	As a user, I want to review my performance across interview sessions so that I can track my progress over time.	3
R16	Review Interview History	As a user, I want to review my previous interview sessions so that I can compare my past practice results.	3

Non-Functional Requirements

Security

- **NF1:** The system shall require users to authenticate before accessing interview history, saved responses, feedback, and progress associated with their account.
- **NF2:** The system shall prevent one authenticated user from accessing another user's saved interview responses, feedback, or progress.

Data Management and Privacy

- **NF3:** The system shall inform the user within 3 seconds of completing the action if collected data is saved to their account.
- **NF4:** The system shall remove user-related data from storage within 24 hours of a confirmed deletion request.
- **NF5:** The system shall preserve saved interview responses, feedback, and progress with 99% availability while the user is logged in.

Usability and Accessibility

- **NF6:** A user shall be able to start a technical mock interview from the main interview interface within three user actions after logging in.
- **NF7:** All primary functions of the application shall be accessible using keyboard navigation without requiring a mouse.
- **NF8:** The system shall provide text-to-speech functionality which conforms with WCAG 2.1 AA audio content requirements.

- **NF9:** The system shall provide adjustable text display options that allow users to change supported text settings, conforming to WCAG 1.4 (Distinguishable).

Cross-Platform Compatibility

- **NF10:** The system shall support the most recent stable versions of Google Chrome, Microsoft Edge, Firefox, and Safari.
- **NF11:** The system shall provide a usable interface on desktop computers, tablets, and mobile devices.

Reliability

- **NF12:** The system shall display an error message when an interview request or AI service request cannot be completed and shall allow the user to retry the operation.

Sponsor Requirements

I have read and approve the material in this document. If there is no external sponsor, the TA or instructor will sign it for accuracy/scope.

Print Name

Signature

Date